

# ArogyaRakshak — Base Paper Identification & Literature Review

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## 1. Base Paper

### 1.1 Selected Base Paper

**Title:** AI in Healthcare: Simplifying Medical Reports for Enhanced Patient Comprehension **Authors:** Rahulprasath S, Pranav Harshan S, Kabilash PV, Lakshithraj A, Sreemathy J **Published in:** 2025 International Conference on Emerging Technologies in Computing and Communication (ETCC), Bangalore, India **DOI:** 10.1109/ETCC65847.2025.11108371 **Publisher:** IEEE

### 1.2 Why This Was Selected as the Base Paper

Three candidate IEEE papers on medical-document AI were reviewed for this role (see §1.4 for the other two and why they were not selected). This one was chosen because it is the closest existing system to ArogyaRakshak's core technical backbone: it combines OCR-based extraction with a RAG-augmented LLAMA model to process medical documents for patient comprehension — architecturally the same combination (extraction → RAG → LLM generation) that underlies BillNyay and SchemeSetu. It is also the most rigorously evaluated of the three, benchmarked on standard medical QA datasets (MedMCQA, PubMedQA, MedQA) rather than qualitative or anecdotal evaluation, which gives ArogyaRakshak's own evaluation section a concrete methodology to follow and improve on.

### 1.3 Base Paper Summary

**Problem addressed:** Medical reports contain complex, jargon-heavy information that is difficult for patients to understand and time-consuming for healthcare providers to summarize.

**Methodology:** The system accepts two input types — unstructured text (patient-described symptoms) and structured documents (scanned medical reports), the latter processed via OCR. Extracted medical concepts are passed to a fine-tuned LLAMA model, augmented with Retrieval-Augmented Generation, which produces a concise, patient-readable summary while pulling in relevant medical knowledge as needed.

**Key results:** The fine-tuned model substantially outperformed the base LLAMA 3 model on standard medical benchmarks — 87.27% vs. 57.82% on MedQA (4-option), and 96.98% vs. 76.23% on clinical-knowledge tasks. This demonstrates that domain-specific fine-tuning combined with retrieval augmentation meaningfully improves medical summarization quality over a generic LLM.

**Limitations relevant to this project (identified through comparison, not stated as criticism of the original work):**

- English-only; no multilingual capability.
- Produces a passive summary as its final output — no actionable next-step document (no appeal, claim, or grievance generation).
- Operates on a single document type per session; no shared context across multiple related documents (a bill, a scheme document, and a prescription are treated independently, not linked).
- No described mechanism for verifying which parts of the generated summary are grounded in retrieved evidence versus generated freely by the LLM — i.e., no explicit hallucination-mitigation step.
- General medical-report domain; not benchmarked against India-specific structured data (government rate schedules, scheme eligibility rules, drug price ceilings).

#### 1.4 Other Papers Considered for the Base Paper Role

**"Medical Report Simplification System... using NLP-based Extraction and AI-Driven Explanation"** (Mariammal, Swetha K., Samuel Jimrys S., ICICV 2025) — uses SciSpaCy + UMLS for term extraction and an LLM for explanation generation. Not selected as base paper because UMLS is an English-only resource with no reported multilingual pathway, and the paper reports no formal benchmark evaluation. Retained as a supporting reference (§2.3).

**"Healthcare Record Optimization using Natural Language Processing and Computer Vision" (MedScribeAI)** (Singh, Mehrotra, Rosaline, GCWCN 2025) — combines Azure OCR with PubMedBERT for prescription digitization, plus a disease-prediction module. Not selected as base paper because its scope (handwritten prescription digitization) is narrower than the whole-project scope needed here, and its diagnostic-prediction component sits outside ArogyaRakshak's deliberately informational (non-diagnostic) design. Retained as a supporting reference for DawaCheck's OCR/NLP extraction methodology only (§2.3) — explicitly excluding its diagnosis-prediction approach.

1.5 How ArogyaRakshak Extends the Base Paper

| Base paper                                                                    | ArogyaRakshak's extension                                                                                                                                                                                    |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single-module medical report summarization                                    | Four linked modules (BillNyay, SchemeSetu, DawaCheck, DaaviSetu) sharing one context layer (Kadi), so one upload informs multiple outputs                                                                    |
| English-only                                                                  | Full trilingual support — English, Hindi, Marathi — via IndicXlit transliteration matching and IndicSBERT cross-lingual embeddings                                                                           |
| Output is a passive summary                                                   | Output is an actionable document — a drafted insurance appeal, a pre-filled claim form, or a grievance complaint ready for the user to submit                                                                |
| No explicit hallucination-mitigation step described                           | Mandatory citation-enforcement: a QA Judge agent verifies each generated claim against a retrieved source before finalizing any document; unsupported claims are dropped                                     |
| Documents processed independently                                             | Kadi's shared entity-resolution pipeline links entities (medicines, procedures, amounts) across a patient's documents, cross-script included                                                                 |
| General medical-report domain, no benchmarked ground truth beyond QA datasets | Outputs grounded against verifiable Indian government reference data — CGHS rate schedules, PMJAY/MJPJAY scheme rules, NPPA drug price ceilings — giving every flagged deviation a citable, checkable source |

This positions ArogyaRakshak as taking the base paper's core technique (OCR + RAG + LLM for patient-facing medical document comprehension) and extending it along three independent axes: **scope** (one document → linked multi-document reasoning), **language** (English-only → trilingual), and **actionability** (summary → submittable legal/financial document), while adding a stronger grounding and hallucination-control mechanism than the base paper describes.

2. Related Work by Topic

2.1 Overcharge / Fraud Detection (BillNyay)

- Fraud detection in healthcare billing and claims (*International Journal of Science and Research Archive*, 2024) — NLP-based detection of discrepancies between narrative descriptions and billed services; general framing reference for the overcharge-detection problem class.
- Improving Medicare Fraud Detection Accuracy through Feature Selection and Data Sampling Techniques (*medRxiv*, 2026) — reports 95.4% detection accuracy

combining feature selection and data sampling; a useful benchmark figure for BillNyay's own evaluation section.

- Dialogue to Detection: A Multimodal Hybrid NLP Pipeline for Insurance Fraud Detection (*arXiv*, 2026) — combines NER, regex extraction, and LLM-RAG retrieval into a rule-based risk score; supports BillNyay's hybrid rules+RAG+agent design as a defensible architectural pattern.
- *Note*: the above target insurer-side fraud detection; BillNyay solves the mirror-image problem (protecting the patient). Cited for methodological parallels, not direct precedent.

## 2.2 Indic Entity Resolution, Transliteration & NER (Kadi)

- Naamapadam: A Large-Scale Named Entity Annotated Data for Indic Languages (*arXiv/ACL*) — large-scale NER dataset across Indic languages; baseline reference for Kadi's entity extraction.
- L3Cube-MahaNER (Litake et al., WILDRE-6 Workshop, LREC 2022) — Marathi-specific NER dataset from the same research group behind the IndicSBERT model already in use.
- Aksharantar: Towards Building Open Transliteration Tools for the Next Billion Users (Madhani et al., 2022) — open transliteration resource underlying the IndicXlit family of tools used in Kadi's transliteration-matching signal.
- Evaluating Cross-lingual Knowledge Consistency in Code-Mixed vis-a-vis Indian Languages using IndicKLAR (*arXiv*, 2026) — documents transliteration failure modes across Hindi and Marathi; supports the design decision to route medium-confidence entity matches to user confirmation rather than auto-merging.

## 2.3 Extraction Methodology (BillNyay / DawaCheck)

- Medical Report Simplification System (Mariammal et al., ICICV 2025) — SciSpaCy + UMLS term extraction and LLM-based explanation generation; precedent for the term-extraction-then-explanation pattern (English-only precedent this project extends to Hindi/Marathi).
- MedScribeAI (Singh et al., GCWCN 2025) — Azure OCR + PubMedBERT pipeline for handwritten prescription digitization; cited for its extraction methodology only, explicitly excluding its diagnosis-prediction component.

## 2.4 Hallucination Mitigation / RAG Grounding (BillNyay's QA Judge agent)

- Reducing Hallucinations in Medical AI Through Citation-Enforced Prompting in RAG Systems (*MDPI*, 2026) — introduces an "Unsupported Sentence Ratio" metric and citation-enforced grounding, closely matching the QA Judge agent's design; also notes multilingual RAG grounding as an open problem.
- MEGA-RAG: mitigating hallucinations of LLMs in public health (*PMC*, 2026) — semantic-divergence and evidence-coverage scoring approach, referenced for the QA Judge's citation-checking logic.

- HalluGraph: Auditable Hallucination Detection for Legal RAG (*arXiv*, 2025/2026) — reports 17–33% hallucination rates on legal-document generation even with RAG; a realistic baseline to benchmark BillNyay's own appeal/complaint drafting against.
- Mitigating Hallucination in Large Language Models: An Application-Oriented Survey on RAG, Reasoning, and Agentic Systems (*arXiv*, 2026) — broad survey anchoring the hallucination-mitigation discussion before the more specific papers above.

## **2.5 ABDM / Digital Health Interoperability (Kadi's ABDM sandbox integration)**

- Awareness of and Challenges in Utilizing the Ayushman Bharat Digital Mission for Healthcare Delivery (*PMC*) — qualitative data on low awareness and data-security concerns as ABDM adoption barriers; motivates why a simple, trustworthy consent flow matters for Kadi's ABDM integration.
- A Review on Interoperability Challenges between EMR (Micro) and ABDM (Macro) (*ResearchGate/IRJCS*) — documents real interoperability challenges, supporting the decision to scope ABDM integration to sandbox-only rather than claim full production interoperability.

## **2.6 OCR — Devanagari / Indic Handwriting (OCR validation methodology)**

- Devanagari Handwritten Character Dataset (DHCD, UCI Machine Learning Repository) — 92,000 images across 46 classes; baseline character-level Devanagari OCR benchmark.
- IIIT-HW-Dev (*IEEE*) — 95,381 word-level Devanagari handwriting samples; closer to real prescription/bill reading than character-level datasets.
- CHIPS — Corpus of Handwritten Indic Scripts — page-level benchmark across 13 Indic languages; the most realistic available precedent for validating OCR on full scanned documents.

## **2.7 RAG for Policy/Legal Eligibility Reasoning (SchemeSetu)**

- A Reasoning-Focused Legal Retrieval Benchmark (*arXiv*, 2025) — benchmarks RAG on eligibility-style statute reasoning where the query and relevant document share little lexical overlap, precisely SchemeSetu's retrieval problem.
  - CBR-RAG: Case-Based Reasoning for RAG in Legal Question Answering (*arXiv*) — proposes case-based retrieval structuring as an alternative to pure semantic retrieval; potential design reference for improving eligibility-answer consistency.
  - Knowledge Graph Representations for LLM-Based Policy Compliance Reasoning (*arXiv*, 2026) — reports reduced hallucination when legal information is knowledge-graph structured rather than flat-text retrieved; noted as a future-work direction for SchemeSetu, not required for the current submission.
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### 3. Base Paper vs. Supporting Literature — Summary

| Category                   | Role in this documentation                                                                                                                                                         |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base paper (§1)            | The single closest prior system; ArogyaRakshak is positioned as a direct extension of its technique along scope, language, and actionability                                       |
| Supporting literature (§2) | Justifies individual design decisions module-by-module; none of these individually cover the project's full scope, which is itself part of the argument for this project's novelty |

### 4. Conclusion

ArogyaRakshak takes the OCR-RAG-LLM approach to patient-facing medical document comprehension established in the base paper and extends it into a linked, trilingual, action-generating system grounded in verifiable Indian government reference data. The supporting literature above demonstrates that each individual component of this extension — Indic entity resolution, citation-enforced hallucination mitigation, Devanagari OCR, and eligibility-focused RAG retrieval — is independently well-studied, while the specific combination attempted here (across four linked modules, in three languages, for Indian healthcare billing/scheme/medicine documents) is not directly addressed by any single existing work.